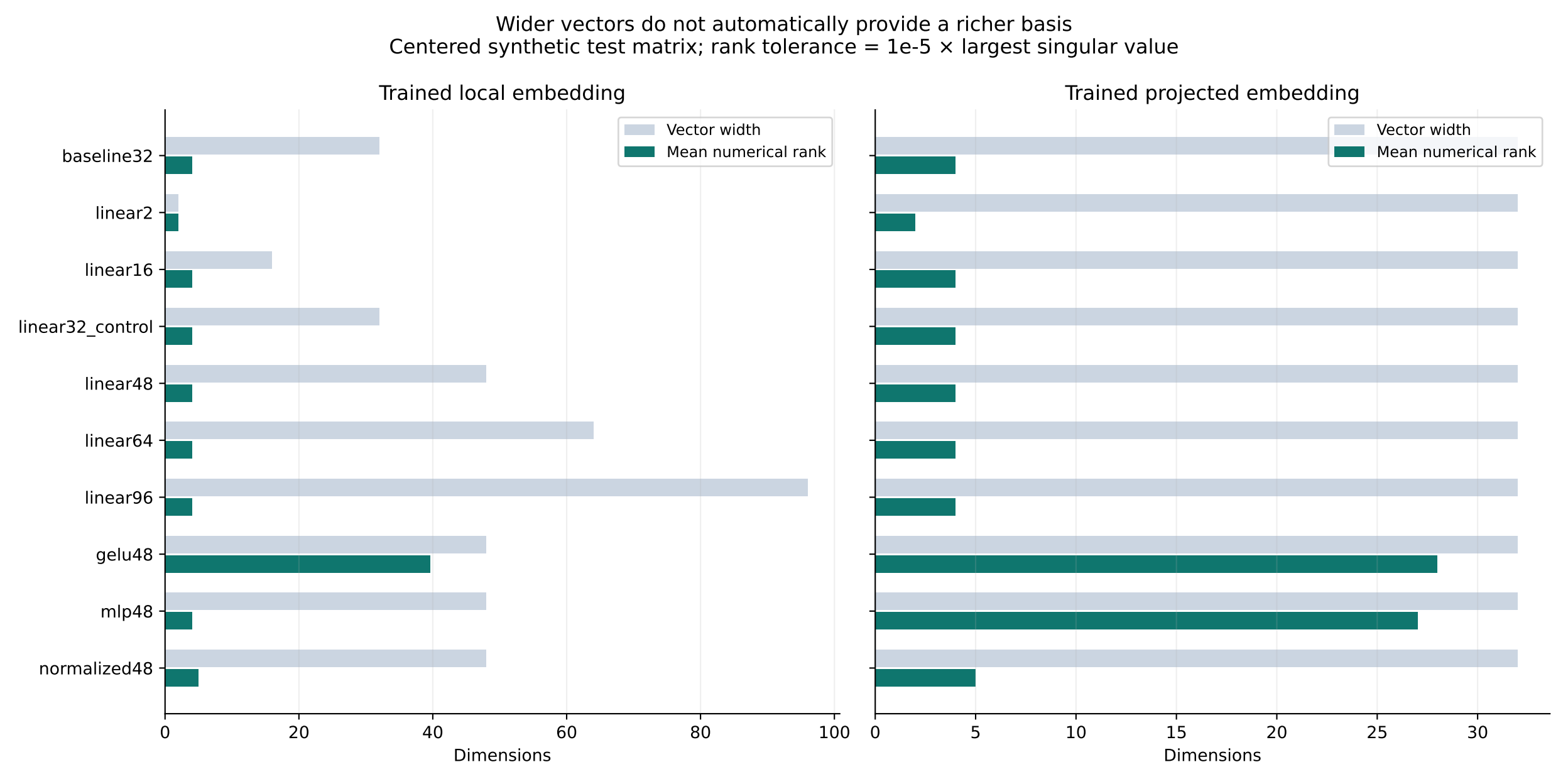
Wider vectors do not automatically provide a richer basis
Centered synthetic test matrix; rank tolerance = 1e-5 × largest singular value
Trained local embedding
Trained projected embedding
Vector width
Mean numerical rank
baseline32
linear2
linear16
linear32_control
linear48
linear64
linear96
gelu48
mlp48
normalized48
0
20
40
60
80
100
0
5
10
15
20
25
30
Dimensions
Dimensions